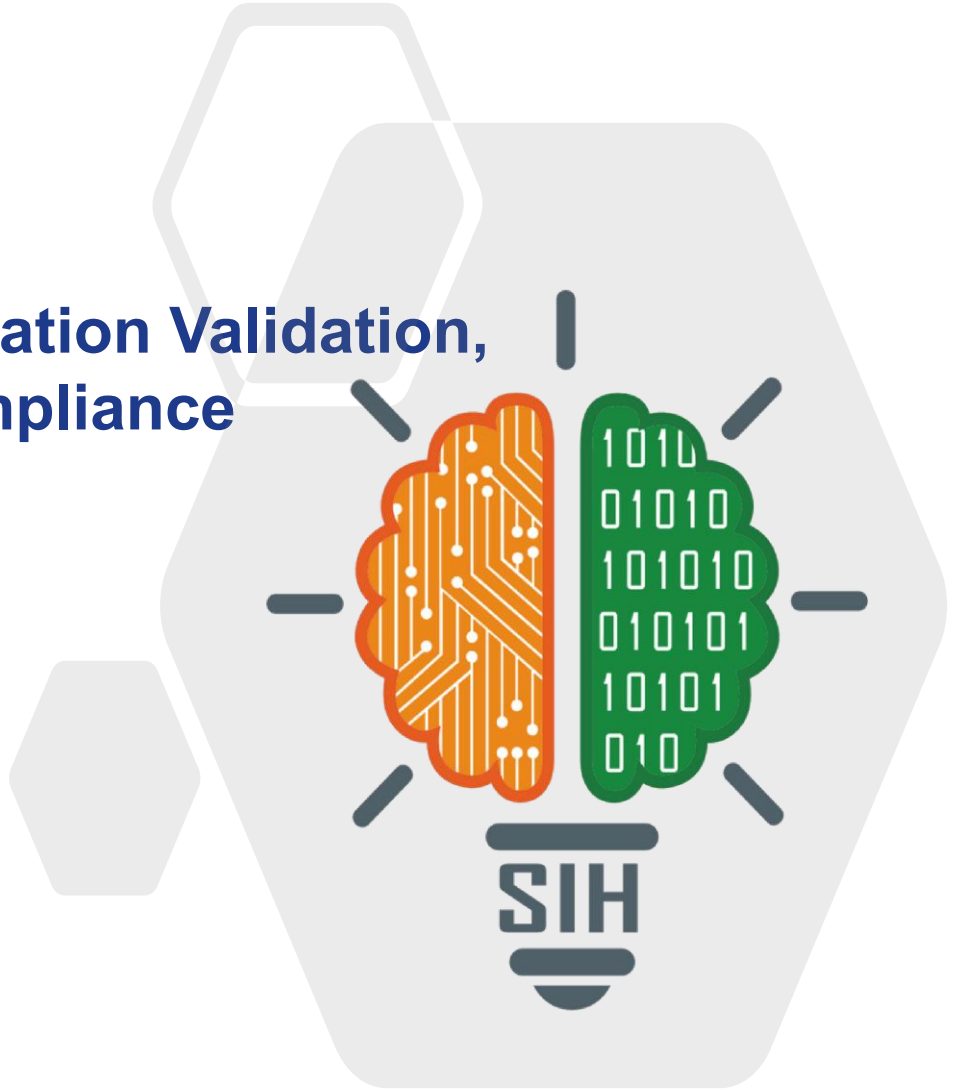
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REGULENS AICTE

- Problem Statement ID – SIH26261
- Problem Statement Title – AI-Powered Application Validation, Document Authentication and Regulatory Compliance Verification
- Theme – Smart Automation
- PS Category – Software
- Team ID – 198271
- Team Name – PROOFX



PROOFX

REGULENS AICTE

"An AI Decision-Support Ecosystem for Institutional Application Validation, Document Forensics & APH Compliance"

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Problems in Real Life

- **Manual scrutiny bottleneck -** 10,000+ colleges submit 300,000+ pages annually, delaying approvals by 3-5 months.
- **Document forgery & alteration -** Photoshops on expired Fire NOCs and fake building safety stamps slip through.
- **Sub-leased / disputed land -** Discrepancies in registered land titles, Khasra numbers, and encumbrances.
- **Inflated carpet area declarations -** Instructional space declared in web forms doesn't match architect blueprints.
- **Ghost faculty duplication -** Identical professors registered on multiple college payrolls simultaneously.

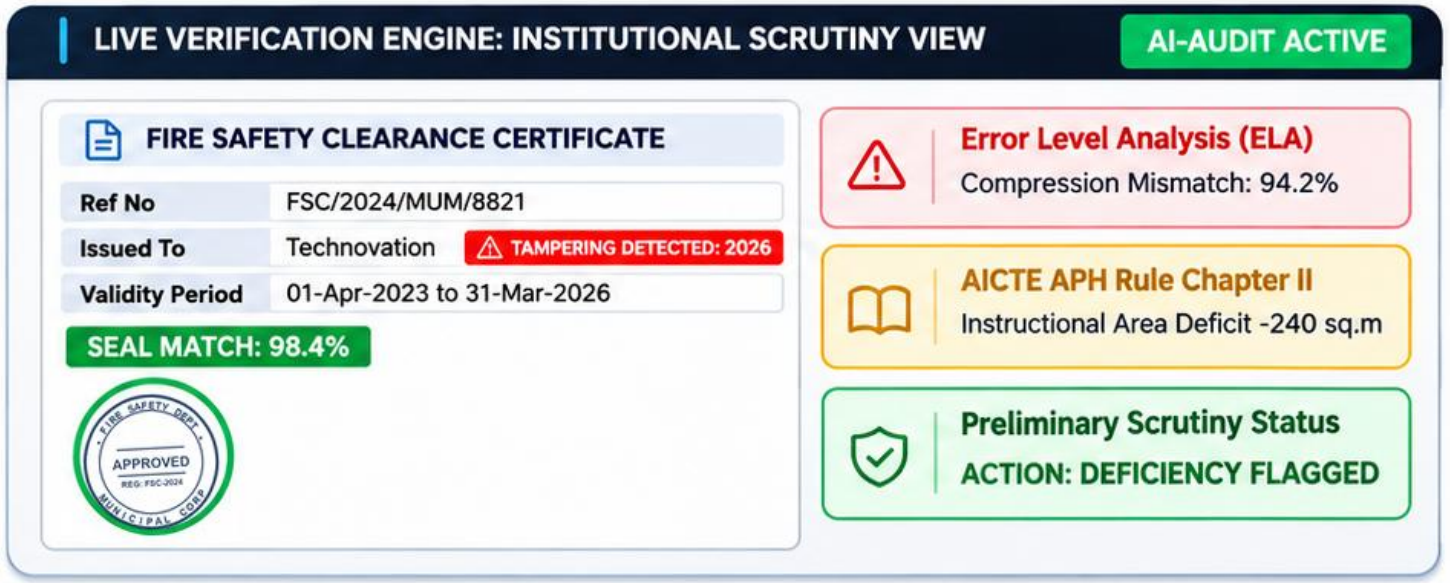
Proposed solution

Upload PDF / Scan → **OCR Engine** PaddleOCR → **Forensics** ELA & EXIF → **APH Rules** Constraint Math → **Cockpit** Officer PSR

- **Multimodal Ingestion :** Statutory PDFs, structural blueprints, and registration registers.
- **Pixel-Level ELA Forensics :** Detects JPEG resave compression differences and fake stamps.
- **Left-Shift Pre-Flight Audit :** Allows institutions to pre-audit before formal submission.
- **Local Events :** Deficiency flags, carpet deficit, and altered date alerts.
- **APH Rule Engine :** Encodes AICTE Approval Process Handbook (APH 2024-27) norms.
- **NO DATA LEAKAGE :** 100% on-premise execution; zero sovereign data leaves government.

Challenges in current system

- **Blind file uploads :** Portals accept raw PDFs without forensic tamper detection or authentic origin validation.
- **Extreme manual fatigue :** Scrutiny officers review 50+ complex legal and structural documents per college manually.
- **Complex mathematical APH rules :** Calculating faculty cadre ratios (1:2:6) and FSI across branches manually causes errors.
- **Multi-document inconsistency :** Information spread across web forms, affidavits, and drawings is rarely reconciled.
- **Lack of verifiable audit trail :** No cryptographic provenance to prevent post-facto alterations or internal collusion.



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TECHNICAL APPROACH

PHASED PIPELINE WITH VERIFICATION LOOP

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Ingest
- Multi-page statutory PDF & Scan
- Adaptive deskewing & binarization
- Frame & layer sampling

→

Detect & Extract
- PaddleOCR + LayoutLMv3
- Table & Key-Value extraction
- Blueprint dimension parsing

→

Forensic Audit
- Error Level Analysis (ELA)
- Metadata / EXIF editing audit
- ORB/SIFT seal template match

→

Decide (Rules)
- AICTE APH 2024-27 norms
- Cadre ratio math (1:2:6)
- Carpet area vs student intake

→

Validate & Act
- Automated PSR dossier
- Officer split-screen review
- SHA-256 cryptographic audit

SYSTEM ARCHITECTURE

INPUT INGESTION
- Land Deeds
- Fire Safety NOC
- Blueprint Layout
- Faculty Rolls
- FDR Receipts

→

REGULENS ENGINE (On-Premise Core)
- PaddleOCR & LayoutLM
- ELA Tamper Detector
- SIFT/ORB Seal Matcher
- APH Symbolic Solver
- SHA-256 Audit Hasher

→

DECISION COCKPIT
- Officer Workbench
- Pre-Flight Portal
- Automated PSR
- AICTE Registry Sync

WHAT THE SCRUTINY OFFICER SEES

Scrutiny Overview: Live institutional compliance score, pending applications in desk queue, and critical deficiency breakdown.

Split-Screen Document Viewer: Original uploaded PDF on the left with glowing red/amber bounding boxes citing exact irregularities.

APH Compliance Checklist: Automated pass/fail evaluation mapped directly to AICTE Approval Process Handbook (APH) clauses.

One-Click PSR Dossier: Official Preliminary Scrutiny Report generated with verifiable cryptographic hash, ready for committee sign-off.

TECH STACK

OpenCV | Layout LMv3 | Paddle OCR | Python | FastAPI | SQLite/ Postgres | React.js | Tailwind

DOCUMENT FORENSICS (ELA)

[Tampered Pixels Detected]
Compression delta: 94.2%
Font Mismatch: Alert

REAL-TIME MODEL RUN

- Land: VERIFIED (Pass)
- Carpet: -240 sq.m (Deficit)
- Cadre 1:2:6: VERIFIED

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FEASIBILITY AND VIABILITY

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Technical Feasibility

- **CPU-friendly:** Core ELA forensics and OpenCV run on standard servers without expensive GPUs.
- **Modular** REST API connects directly to AICTE portal without DB changes.
- **Offline capability:** Scrutiny engine functions even without Internet.

Operational Feasibility

- **Easy Integration:** Fits institutional pre-flight workflow without disrupting colleges.
- **Automatic Scrutiny:** Pre-populates PSR report for officers.
- **Human-in-the-Loop (HITL):** AI assists; final statutory sign-off remains with human officers.

Feasibility Analysis and viability

Economic Feasibility

- **Lower Setup Cost:** 100% open-source software (OpenCV, FastAPI, React) with zero vendor license fees.
- **Reduced Operating Cost:** Cuts manual scrutiny hours by 75% and eliminates wasteful Expert Visit travel.

Scalability Feasibility

- **Start Small:** Pilot across 2 AICTE regional offices and expand nationally.
- **Modular Expansion:** Land, Building, Fire NOC, and Faculty modules can be deployed independently.

Risk Management for Regulatory AI Systems

False Alerts

- Human Verification
- Use Temporal Validation

Poor Scan Quality

- Adaptive Gaussian Binarization
- Deskewing & Noise Filter

Portal API Failures

- Standard REST Interfaces
- Modular Connectors

Risk Management for Regulatory AI

Internet Outage

- Offline Edge AI Processing
- Sync When Reconnected

Hardware Failures

- Health Monitoring
- Failover & Local Backup

Privacy / Forgery

- Sovereign On-Premise
- SHA-256 Audit Ledger

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IMPACT AND BUSINESS MODEL

ReguLens AICTE

SMART INDIA HACKATHON 2026

IMPACT

- **Operational Efficiency :** 75% reduction in scrutiny turnaround time (5 mins per college vs 4 hours); eliminates backlog.
- **Accreditation Integrity :** Prevents fraudulent colleges operating with doctored Fire NOCs or fake architect certificates.
- **Student & Parent Safety :** Ensures zero student admissions into sub-standard, safety-compromised, or illegal technical colleges.
- **National Tech Modernization :** Proves light-but-tight digital regulation model for NEP 2020 across UGC, PCI, and BCI.

AICTE REGULATORY COMPLIANCE RADAR



BUSINESS / SUSTAINABILITY MODEL

CUSTOMERS / USERS

- AICTE Regulation Bureau & Regional Scrutiny Cells
- 10,000+ Technical Institutions & Universities

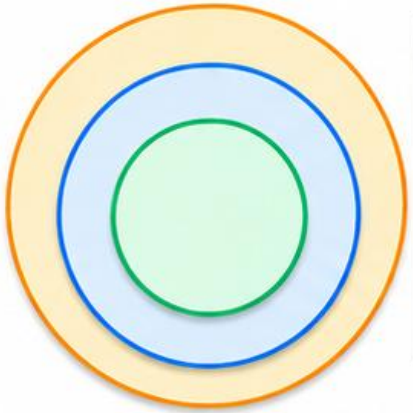
COST RECOVERY

- Nominal self-audit fee per college during EoA
- Replaces ₹40+ Crores spent on manual scrutiny committees

VALUE PROPOSITION

- Sub-minute turnaround
- 100% privacy-safe on-premise
- Cryptographic tamper-evidence

MARKET SIZE (INSTITUTIONAL SCALE)

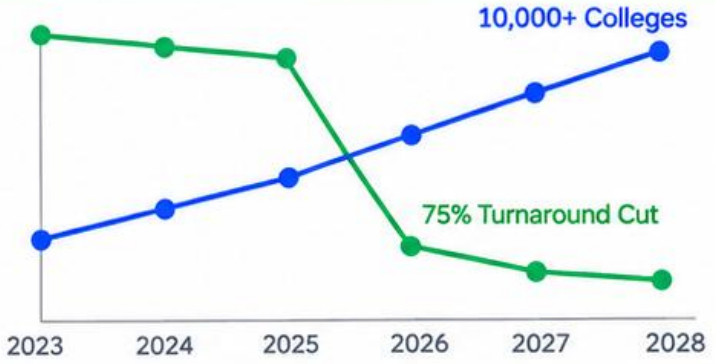


TAM: 50,000+
Higher Ed Institutes (UGC + AICTE + State)

SAM: 10,000+
AICTE Colleges

SOM: ~2,500
Year 1 Target

AICTE REGULATORY LOAD & DIGITAL TRANSFORMATION



Source: AICTE Annual Reports & SIH Projections

AICTE REGULENS COCKPIT

SCRUTINY DESK

Compliance Score
92.4%

Active Desk Queue
12 Pending
Avg Turnaround: 3.8 mins

AICTE Approval Handbook Verification Status

Land: Verified | Carpet Deficit | Fire NOC: Altered

GENERATE OFFICIAL PSR REPORT

PROOFX

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2026

TREND: INDIA TECHNICAL EDUCATION ACCREDITATION

10,000+
Colleges regulated by AICTE across India

300,000+
Annual scrutiny pages verified in EoA cycle

75% Cut
In regulatory turnaround time achieved by AI

COMPETITIVE LANDSCAPE

Solution	Approach	Gap / ReguLens Advantage
Manual Scrutiny Committees	Paper / PDF manual reading	3–5 months delay; misses subtle digital edits & fake stamps.
Current AICTE Web Portal	Blind document upload forms	Zero document authenticity checks; no cross-doc validation.
Commercial OCR Tools (ABBYY)	Generic text extraction	No APH regulatory rule logic; high commercial licensing cost.
Smart CCTV / People Counters	Physical site inspection only	Requires expensive physical visit; post-facto detection.
ReguLens AICTE (Our Solution)	**Multimodal Forensics + APH Rule Engine**	**Sub-minute automated PSR; pixel tamper detection; open-source.**

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